

Dynamics in the center manifold of the collinear points of the restricted three body problem

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Abstract

This paper focuses on the dynamics near the collinear equilibrium points $L_{1,2,3}$ of the spatial Restricted Three Body Problem (RTBP). It is well known that the linear behavior of these three points is of the type center $\times$ center $\times$ saddle. To obtain an accurate description of the dynamics in an extended neighborhood of those points, two different (but complementary) strategies are used. First, the Hamiltonian of the problem is expanded in power series around the equilibrium point. Then, a partial normal form scheme is applied in order to uncouple (up to high order) the hyperbolic directions from the elliptic ones. Skipping the remainder we have that the (truncated) Hamiltonian has an invariant manifold tangent to the central directions of the linear part. The restriction of the Hamiltonian to this manifold is the so-called reduction to the center manifold. The study of the dynamics of this reduced Hamiltonian (now with only two degrees of freedom) gives a qualitative description of the phase space near the equilibrium point. Finally, a Lindstedt–Poincaré procedure is applied to explicitly compute the invariant tori contained in the center manifold. These tori are obtained as the Fourier series of the corresponding solutions, the frequencies being a power expansion of some parameters (amplitudes). This allows for an accurate quantitative description of these regions. In particular, the well known Halo orbits are obtained. ©1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

Let us start with a brief description of the so-called Restricted Three Body Problem (from now on, RTBP). Consider the motion of an infinitesimal particle under the gravitational attraction (Newton's law) of two point like masses called primaries. The attraction of the infinitesimal particle on the primaries has been neglected (as if it had zero mass) so the primaries are describing Keplerian orbits around their common center of masses. Then, the study of the motion of the infinitesimal particle is what is known as RTBP. Here we will assume (without further mention)

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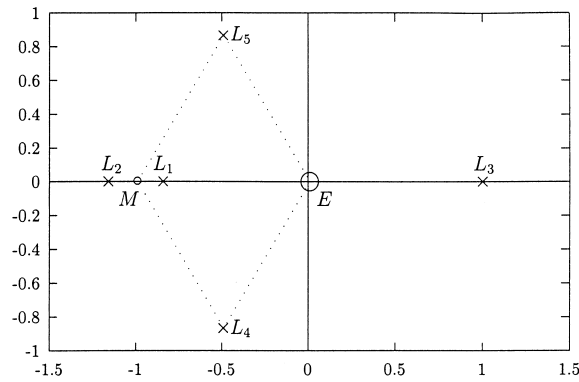


Fig. 1. The five equilibrium points of the RTBP.

that the primaries are moving on a circular orbit, although the case in which they move on an elliptic trajectory is also very relevant. The RTBP is used as a first model to study several problems of Celestial Mechanics. For instance, to study the motion of an asteroid under the attraction of Jupiter and Sun, or the motion of an artificial satellite in the Earth–Moon system.

To simplify the equations of motion, let us take units of mass, length and time such that the sum of masses of the primaries, the gravitational constant and the period of the motion of the primaries is 1, 1 and 2π respectively. With these units the distance between the primaries is also equal to 1. We denote by μ the mass of the smallest primary (the mass of the biggest one is then $1 - \mu$), $\mu \in (0, 1/2]$. A usual system of reference (called synodical system) is the following: the origin is taken at the center of masses of the two primaries, the X -axis is given by the line that goes from the smallest to the biggest primary (and with this orientation), the Z -axis has the direction given by the angular motion of the primaries (and with the same orientation) and the Y -axis is chosen to have a positively oriented system of reference. Hence, in the synodical system the primary of mass μ is always located at $(\mu - 1, 0, 0)$ and the primary of mass $1 - \mu$ at $(\mu, 0, 0)$.

Defining momenta as $P_X = \dot{X} - Y$, $P_Y = \dot{Y} + X$ and $P_Z = \dot{Z}$, the equations of motion can be written in Hamiltonian form. The corresponding Hamiltonian function is (see [1])

$$H = \frac{1}{2}(P_X^2 + P_Y^2 + P_Z^2) + YP_X - XP_Y - \frac{1 - \mu}{r_1} - \frac{\mu}{r_2}. \quad (1)$$

It is also well-known that the RTBP in synodical coordinates has five equilibrium points. Two of them can be found as the third vertex of the two equilateral triangles that can be formed using the two primaries as vertices (they are also called triangular points, Lagrangian points or simply $L_{4,5}$). The collinear points lie on the X -axis and are also called $L_{1,2,3}$ or Eulerian points (see Fig. 1).

In this paper we are going to focus on the dynamics near the collinear points. The linearized vectorfield at these points exhibits a behavior of the kind center $\times$ center $\times$ saddle (for all μ), so these equilibrium points are unstable. On the other hand, there exist solutions that are always close to them, such as the two families of periodic Lyapunov orbits related to the two linear oscillators of the linearized vectorfield. As one of these families is contained in the horizontal plane and the other one is tangent to the (Z, P_Z) direction at $L_{1,2,3}$, we will refer to them as the planar and vertical (Lyapunov) families, respectively. Other interesting solutions are the two-dimensional invariant tori related to the Lissajous solutions, obtained as the product of the two linear oscillations (see [2,3] for proofs of the existence of such tori), and the Halo orbits, obtained as a bifurcation of the planar family of Lyapunov orbits.

In order to describe the dynamics in a relatively big neighborhood of the collinear points, we will perform the so-called reduction to the center manifold. This process is based on expanding the initial Hamiltonian around a given

equilibrium point and performing a partial normal form scheme, uncoupling (up to a high order) the hyperbolic directions from the elliptic ones. The restriction to the invariant manifold tangent to these elliptic directions is a two degrees of freedom Hamiltonian system with an elliptic equilibrium point at the origin. The study of the dynamics of this Hamiltonian is done by fixing an energy level (the phase space is now 3-D) and doing a suitable (2-D) Poincaré section, that can be easily plotted. In this way, varying the value of the energy, we will obtain a sequence of 2-D plots describing the dynamics contained in the (approximately invariant) center manifold.

Once the dynamics has been qualitatively explained, we face the problem of computing accurately some of the solutions of the center manifold. To this end we discuss the Lindstedt–Poincaré procedure (see [4]), that is directly applied to the initial equations (the ones related to (1)). This process is based on finding a parametric family of trigonometric expansions that satisfy the equations of motion, up to a sufficiently high order. Once these expansions have been found, it is very easy to plot periodic and quasi-periodic solutions contained in the center manifold.

The advantage of the Lindstedt–Poincaré method is that it produces explicit formulae for the solutions in the initial coordinates, so that we obtain a (single) compact parametric expression for all the trajectories, whose parameters are the amplitudes for each frequency (see Section 4.1). The main disadvantage for our purpose here is that a bifurcation (or a separatrix) cannot be contained in such expressions, so we cannot expect to detect such phenomena by computing the Lindstedt–Poincaré series. On the other hand, the reduction to the center manifold used here contains the main bifurcation present in the problem (the one giving rise to the Halo orbits), so we can describe the dynamics across this bifurcation. Note that using a full normalization we can also describe the dynamics close to the origin, but we cannot ‘go through’ the bifurcation. Moreover, an integrable normal form is not the best method to compute quasi-periodic solutions that are close to the origin. Although it is easy to write a closed expression for the solution in normal form coordinates, to send this expression back to the initial ones is a relatively big task. One can also send not the complete expression but simply some points. Even in this case, the cost of the transformation for each point is bigger than the cost using the Lindstedt–Poincaré series.

Since 1978, when NASA launched the ISEE-3 spacecraft, Lissajous and Halo type trajectories around the collinear equilibrium points have been considered in the trajectory design of space missions. Since the end of 1996 the SOHO spacecraft is using a Halo orbit around L_1 in the Earth–Sun system as a nominal station orbit. Moreover, in the near future more complex missions to the same location are envisioned, such as the GENESIS mission, planned to be launched by the year 2001 (see also [5–10]). Semi-analytical expansions up to a high order, like the ones obtained by the Lindstedt–Poincaré procedure, can give accurate trajectories for the RTBP model, that can be easily improved for more realistic models for the motion of the spacecraft in the solar system [11,12]. The RTBP solution is, in many cases, only slightly deformed when it is recomputed for a realistic model (under some extra hypotheses, Ref. [13] contains some theoretical results about the effect of the main perturbations of these kind of problems on previously existing periodic and quasi-periodic solutions). Hence, the knowledge of all the librating trajectories given by the reduction to the center manifold allows to select a very suitable approximation for a given purpose. We note the flexibility of this design process, simplifying the study of complex missions and giving a fast adaptation to the technical requirements.

Let us remark that all the computations mentioned here have been done using algebraic manipulators written from scratch (in C and Fortran) by the authors, taking advantage of the particularities of the problem. This, jointly with the use of double precision coefficients for the involved expansions, allows to obtain the results presented here within a reasonable amount of computer time and memory. A general purpose algebraic manipulator is not efficient enough to produce these results using actually existing computers (see also [14]).

Finally, let us mention that the computation of the center manifold to obtain a complete description of the dynamics was first considered (as far as we know) in [11], in order to describe a big neighborhood of the L_1 point of the Earth–Sun system. Related computations can be found in [15–17]. The Lindstedt–Poincaré method is first described in [4], and it has been used in many contexts (see, for instance, [18] or [11]).

The paper has been organized as follows: In Section 2 we introduce the equations of motion as well as the expansions used; Section 3 is devoted to the reduction to the center manifold and Section 4 contains the details of the computation of invariant tori near the collinear equilibrium points. A discussion of the results is also contained in Sections 3 and 4.

2. Expansion of the equations of motion

The equations of motion corresponding to (1) are usually written as

$$\ddot{X} - 2\dot{Y} = \frac{\partial \Omega}{\partial X}, \quad \ddot{Y} + 2\dot{X} = \frac{\partial \Omega}{\partial Y}, \quad \ddot{Z} = \frac{\partial \Omega}{\partial Z}, \quad (2)$$

where $\Omega = (1/2)(X^2 + Y^2) + (1 - \mu)/r_1 + \mu/r_2$. Let us start by translating the origin of coordinates to the selected point $L_{1,2,3}$. It is well known [1] that the distance from L_j to the closest primary, γ_j , is given by the only positive solution of the Euler quintic equation,

$$\begin{aligned} \gamma_j^5 \mp (3 - \mu)\gamma_j^4 + (3 - 2\mu)\gamma_j^3 - \mu\gamma_j^2 \pm 2\mu\gamma_j - \mu &= 0, \quad j = 1, 2, \\ \gamma_j^5 + (2 + \mu)\gamma_j^4 + (1 + 2\mu)\gamma_j^3 - (1 - \mu)\gamma_j^2 - 2(1 - \mu)\gamma_j - (1 - \mu) &= 0, \quad j = 3, \end{aligned}$$

where the upper sign in the first equation is for L_1 and the lower one for L_2 . These equations can be solved numerically by the Newton method, using as starting point $(\mu/3)^{1/3}$ for the first equation ($L_{1,2}$ cases), and $1 - (7/12)\mu$ for the second one (L_3 case).

In order to have good numerical properties for the coefficients of the Taylor expansion it is very convenient to introduce some scaling (see [19]). The translation to the equilibrium point plus the scaling is given by

$$X = \mp \gamma_j x + \mu + a, \quad Y = \mp \gamma_j y, \quad Z = \gamma_j z,$$

where the upper sign corresponds to $L_{1,2}$, the lower one to L_3 , $a = -1 + \gamma_1$ for L_1 , $a = -1 - \gamma_2$ for L_2 and $a = \gamma$ for L_3 . Note that this change redefines the unit of distance as the distance from the equilibrium point to the closest primary. As scalings are not canonical transformations, they have to be applied on the equations of motion (2). In order to expand the nonlinear terms, we will use that

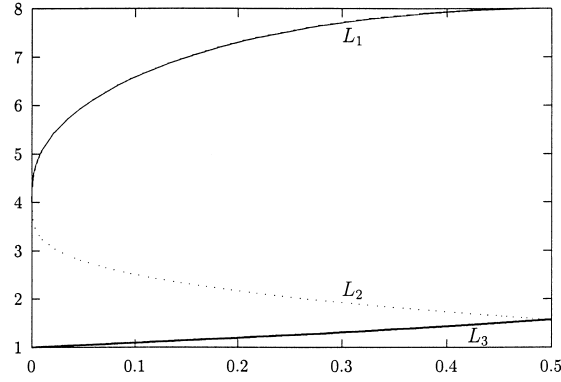
$$\frac{1}{\sqrt{(x - A)^2 + (y - B)^2 + (z - C)^2}} = \frac{1}{D} \sum_{n=0}^{\infty} \left(\frac{\rho}{D}\right)^n P_n \left(\frac{Ax + By + Cz}{D\rho}\right),$$

where $D^2 = A^2 + B^2 + C^2$, $\rho^2 = x^2 + y^2 + z^2$ and P_n is the Legendre polynomial of degree n . After some calculations, one obtains that the equations of motion can be written as

$$\begin{aligned} \ddot{x} - 2\dot{y} - (1 + 2c_2)x &= \frac{\partial}{\partial x} \sum_{n \geq 3} c_n(\mu) \rho^n P_n \left(\frac{x}{\rho}\right), \quad \ddot{y} + 2\dot{x} + (c_2 - 1)y = \frac{\partial}{\partial y} \sum_{n \geq 3} c_n(\mu) \rho^n P_n \left(\frac{x}{\rho}\right), \\ \ddot{z} + c_2 z &= \frac{\partial}{\partial z} \sum_{n \geq 3} c_n(\mu) \rho^n P_n \left(\frac{x}{\rho}\right), \end{aligned} \quad (3)$$

where the left-hand side contains the linear terms and the right-hand side contains the nonlinear ones. The coefficients $c_n(\mu)$ are given by

$$c_n(\mu) = \frac{1}{\gamma_j^3} \left((\pm 1)^n \mu + (-1)^n \frac{(1 - \mu)\gamma_j^{n+1}}{(1 \mp \gamma_j)^{n+1}} \right), \quad \text{for } L_j, j = 1, 2$$


 Fig. 2. Values of $c_2(\mu)$, $\mu \in [0, \frac{1}{2}]$, for the cases $L_{1,2,3}$.

$$c_n(\mu) = \frac{(-1)^n}{\gamma_3^3} \left(1 - \mu + \frac{\mu \gamma_3^{n+1}}{(1 + \gamma_3)^{n+1}} \right), \quad \text{for } L_3.$$

As usual, in the first equation, the upper sign is for L_1 and the lower one for L_2 . Note that these equations can be written in Hamiltonian form, by defining the momenta $p_x = \dot{x} - y$, $p_y = \dot{y} + x$ and $p_z = \dot{z}$. The corresponding Hamiltonian is then given by

$$H = \frac{1}{2} (p_x^2 + p_y^2 + p_z^2) + y p_x - x p_y - \sum_{n \geq 2} c_n(\mu) \rho^n P_n \left(\frac{x}{\rho} \right). \quad (4)$$

The nonlinear terms of this Hamiltonian can be expanded by means of the well-known recurrence of the Legendre polynomials P_n . For instance, if we define

$$T_n(x, y, z) = \rho^n P_n \left(\frac{x}{\rho} \right), \quad (5)$$

then, it is not difficult to check that T_n is a homogeneous polynomial of degree n that satisfies the recurrence

$$T_n = \frac{2n-1}{n} x T_{n-1} - \frac{n-1}{n} (x^2 + y^2 + z^2) T_{n-2}, \quad (6)$$

starting with $T_0 = 1$ and $T_1 = x$.

2.1. Preliminaries for the center manifold

The linearization around the equilibrium point is given by the second order terms (linear terms must vanish) of the Hamiltonian that, after some rearranging, takes the form,

$$H_2 = \frac{1}{2} (p_x^2 + p_y^2) + y p_x - x p_y - c_2 x^2 + \frac{c_2}{2} y^2 + \frac{p_z^2}{2} + \frac{c_2}{2} z^2. \quad (7)$$

It is not difficult to derive intervals for the values of c_2 when $\mu \in [0, 1/2]$ (see Fig. 2). As $c_2 > 0$ (for the three collinear points), the vertical direction is an harmonic oscillator with frequency $\omega_2 = \sqrt{c_2}$. As the vertical direction is already uncoupled from the planar ones, in what follows we will focus on the planar directions, i.e.,

$$H_2 = \frac{1}{2} (p_x^2 + p_y^2) + y p_x - x p_y - c_2 x^2 + \frac{c_2}{2} y^2, \quad (8)$$

where, for simplicity, we keep the name H_2 for the Hamiltonian.

The next step will be to compute a symplectic change of variable such that Hamiltonian (8) takes a simpler form, suitable to start the normal form computations. To this end, let us define the 4×4 matrix J as

$$J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix},$$

where I_2 denotes the 2×2 identity matrix. The equations of motion of (8) are given by the linear system

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{p}_x \\ \dot{p}_y \end{pmatrix} = J \nabla H_2 = J \text{Hess}(H_2) \begin{pmatrix} x \\ y \\ p_x \\ p_y \end{pmatrix}.$$

Let us define the matrix M as $J \text{Hess}(H_2)$,

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 2c_2 & 0 & 0 & 1 \\ 0 & -c_2 & -1 & 0 \end{pmatrix}. \quad (9)$$

The characteristic polynomial is $p(\lambda) = \lambda^4 + (2 - c_2)\lambda^2 + (1 + c_2 - 2c_2^2)$. Calling $\eta = \lambda^2$, we have that the roots of $p(\lambda) = 0$ are given by

$$\eta_1 = \frac{c_2 - 2 - \sqrt{9c_2^2 - 8c_2}}{2}, \quad \eta_2 = \frac{c_2 - 2 + \sqrt{9c_2^2 - 8c_2}}{2}.$$

As $\mu > 0$, we have that $c_2 > 1$ that forces $\eta_1 < 0$ and $\eta_2 > 0$. This shows that the equilibrium point is a center $\times$ center $\times$ saddle. Thus, let us define ω_1 as $\sqrt{-\eta_1}$ and λ_1 as $\sqrt{\eta_2}$. For the moment, we do not specify the sign taken for each square root (this will be discussed later on).

Now, we want to find a symplectic linear change of variables casting (8) into its real normal form and, hence, we will look for the eigenvectors of matrix (9). We will take advantage of the special form of this matrix: if we denote by M_λ the matrix $M - \lambda I_4$, then

$$M_\lambda = \begin{pmatrix} A_\lambda & I_2 \\ B & A_\lambda \end{pmatrix}, \quad A_\lambda = \begin{pmatrix} -\lambda & 1 \\ -1 & -\lambda \end{pmatrix}, \quad B = \begin{pmatrix} 2c_2 & 0 \\ 0 & -c_2 \end{pmatrix}.$$

Now, the kernel of M_λ can be found as follows: denoting as $(w_1^\top, w_2^\top)^\top$ the elements of the kernel, we start solving $(B - A_\lambda^2)w_1 = 0$ and then $w_2 = -A_\lambda w_1$. Thus, the eigenvectors of M are given by $(2\lambda, \lambda^2 - 2c_2 - 1, \lambda^2 + 2c_2 + 1, \lambda^3 + (1 - 2c_2)\lambda)^\top$, where λ denotes one of the eigenvalues.

Let us start considering the eigenvectors related to ω_1 . From $p(\lambda) = 0$, we obtain that ω_1 verifies

$$\omega_1^4 - (2 - c_2)\omega_1^2 + (1 + c_2 - 2c_2^2) = 0.$$

We also apply $\lambda = \sqrt{-1}\omega_1$ to the expression of the eigenvector and, separating real and imaginary parts as $u_{\omega_1} + \sqrt{-1}v_{\omega_1}$ we obtain

$$\begin{aligned} u_{\omega_1} &= (0, -\omega_1^2 - 2c_2 - 1, -\omega_1^2 + 2c_2 + 1, 0)^\top, \\ v_{\omega_1} &= (2\omega_1, 0, 0, -\omega_1^3 + (1 - 2c_2)\omega_1)^\top. \end{aligned}$$

Now, let us consider the eigenvalues related to $\pm\lambda_1$,

$$\begin{aligned} u_{+\lambda_1} &= (2\lambda_1, \lambda_1^2 - 2c_2 - 1, \lambda_1^2 + 2c_2 + 1, \lambda_1^3 + (1 - 2c_2)\lambda_1)^\top, \\ v_{-\lambda_1} &= (-2\lambda_1, \lambda_1^2 - 2c_2 - 1, \lambda_1^2 + 2c_2 + 1, -\lambda_1^3 - (1 - 2c_2)\lambda_1)^\top. \end{aligned}$$

We consider, initially, the change of variables $C = (u_{+\lambda_1}, u_{\omega_1}, v_{-\lambda_1}, v_{\omega_1})$. To know whether this matrix is symplectic or not, we check $C^\top J C = J$. It is tedious but not difficult to see that

$$C^\top J C = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}, \quad D = \begin{pmatrix} d_{\lambda_1} & 0 \\ 0 & d_{\omega_1} \end{pmatrix}.$$

This implies that we need to apply some scaling on the columns of C in order to have a symplectic change. The scaling is given by the square root of the factors

$$d_{\lambda_1} = 2\lambda_1((4 + 3c_2)\lambda_1^2 + 4 + 5c_2 - 6c_2^2), \quad d_{\omega_1} = \omega_1((4 + 3c_2)\omega_1^2 - 4 - 5c_2 + 6c_2^2).$$

Thus, we define $s_1 = \sqrt{d_{\lambda_1}}$ and $s_2 = \sqrt{d_{\omega_1}}$. As we want the change to be real, we have to require $d_{\lambda_1} > 0$ and $d_{\omega_1} > 0$. It is not difficult to check that this condition is satisfied for $0 < \mu \leq 1/2$ in all the points $L_{1,2,3}$, provided that λ_1 and ω_1 have been selected with positive sign.

To obtain the final change, we have to take into account the vertical direction (z, p_z) : to put it into real normal form we use the substitution

$$z \rightarrow \frac{1}{\sqrt{\omega_2}}z, \quad p_z \rightarrow \sqrt{\omega_2}p_z.$$

This implies that the final change is given by the symplectic matrix

$$C = \begin{pmatrix} \frac{2\lambda_1}{s_1} & 0 & 0 & \frac{-2\lambda_1}{s_1} & \frac{2\omega_1}{s_2} & 0 \\ \frac{\lambda_1^2 - 2c_2 - 1}{s_1} & \frac{-\omega_1^2 - 2c_2 - 1}{s_2} & 0 & \frac{\lambda_1^2 - 2c_2 - 1}{s_1} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{\omega_2}} & 0 & 0 & 0 \\ \frac{\lambda_1^2 + 2c_2 + 1}{s_1} & \frac{-\omega_1^2 + 2c_2 + 1}{s_2} & 0 & \frac{\lambda_1^2 + 2c_2 + 1}{s_1} & 0 & 0 \\ \frac{\lambda_1^3 + (1 - 2c_2)\lambda_1}{s_1} & 0 & 0 & \frac{-\lambda_1^3 - (1 - 2c_2)\lambda_1}{s_1} & \frac{-\omega_1^3 + (1 - 2c_2)\omega_1}{s_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \sqrt{\omega_2} \end{pmatrix}, \quad (10)$$

that casts Hamiltonian (7) into its real normal form,

$$H_2 = \lambda_1 x p_x + \frac{\omega_1}{2}(y^2 + p_y^2) + \frac{\omega_2}{2}(z^2 + p_z^2), \quad (11)$$

where, for simplicity, we have kept the same name for the variables. To simplify the computations, we have used a complex normal form for H_2 because this allows to solve very easily the homological equations that determine the generating functions used during the computations of the center manifold (see Section 3.1). This complexification

is given by

$$\begin{aligned} x &= q_1, & y &= \frac{q_2 + \sqrt{-1}p_2}{\sqrt{2}}, & z &= \frac{q_3 + \sqrt{-1}p_3}{\sqrt{2}}, & p_x &= p_1, & p_y &= \frac{\sqrt{-1}q_2 + p_2}{\sqrt{2}}, \\ p_z &= \frac{\sqrt{-1}q_3 + p_3}{\sqrt{2}}, \end{aligned} \quad (12)$$

and it puts (11) into its complex normal form,

$$H_2 = \lambda_1 q_1 p_1 + \sqrt{-1} \omega_1 q_2 p_2 + \sqrt{-1} \omega_2 q_3 p_3, \quad (13)$$

being λ_1 , ω_1 and ω_2 real (and positive) numbers.

2.2. Preliminaries for the Lindstedt–Poincaré method

For the computation of the invariant tori we use a Lindstedt–Poincaré method. As this requires to substitute trigonometric expansions into the right-hand side of (3), we will use an iterative expression for these nonlinear terms. From (5) we define

$$R_{n-1}(x, y, z) = \frac{1}{y} \frac{\partial T_{n+1}}{\partial y}.$$

Obviously, R_{n-1} is a polynomial of degree $n - 1$. Moreover we have the identities

$$R_{n-1}(x, y, z) = \frac{1}{z} \frac{\partial T_{n+1}}{\partial z}, \quad \frac{\partial T_{n+1}}{\partial z} = (n+1)T_n.$$

So, the equations of motion can be written as

$$\begin{aligned} \ddot{x} - 2\dot{y} - (1 + 2c_2)x &= \sum_{n \geq 2} c_{n+1}(n+1)T_n, & \ddot{y} + 2\dot{x} + (c_2 - 1)y &= y \sum_{n \geq 2} c_{n+1}R_{n-1}, \\ \ddot{z} + c_2 z &= z \sum_{n \geq 2} c_{n+1}R_{n-1}. \end{aligned} \quad (14)$$

Note that to substitute an expansion into the right-hand side of (14), one can substitute into the recurrence satisfied by those terms: for the T_n one can use (6) and, for the R_n , it is not difficult to see that

$$R_n(x, y, z) = \frac{2n+3}{n+2}xR_{n-1} - \frac{2n+2}{n+2}T_n - \frac{n+1}{n+2}\rho^2 R_{n-2},$$

where $R_0 = -1$ and $R_1 = -3x$. The expansions that appear in this section have already been used in [18].

3. Reduction to the center manifold

The process of reduction to the center manifold is very similar to a normal form calculation. It is based on removing some monomials in the expansion of the Hamiltonian, in order to produce an invariant manifold tangent to the elliptic directions of H_2 . Let us recall that, if $F(q, p)$ and $G(q, p)$ are two functions (where, as usual, q denotes the positions and p the momenta), their Poisson bracket is defined as

$$\{F, G\} = \frac{\partial F}{\partial q} \frac{\partial G}{\partial p} - \frac{\partial F}{\partial p} \frac{\partial G}{\partial q}.$$

In what follows, we will use the following notation. If $x = (x_1, \dots, x_n)$ is a vector of complex numbers and $k = (k_1, \dots, k_n)$ is an integer vector, we denote by x^k the value $x_1^{k_1} \dots x_n^{k_n}$ (in this context we define 0^0 as 1). Moreover, we define $|k|$ as $\sum_j |k_j|$.

3.1. The Lie series method

Let us start by expanding the initial Hamiltonian around the equilibrium point, in the complex coordinates for which the second degree terms are in diagonal form (see Section 2.1). This expansion can be obtained by substituting the linear change obtained from the composition of (10) and (12) into the recurrence (6), that is then applied to compute the last sum in (4). The second degree terms in (4) that are not in the sum are obtained by direct substitution. In this way, the Hamiltonian takes the form

$$H(q, p) = H_2(q, p) + \sum_{n \geq 3} H_n(q, p), \quad (15)$$

where H_2 is given in (13) and H_n denotes an homogeneous polynomial of degree n .

The changes of variables are implemented by means of the Lie series method: if $G(q, p)$ is a Hamiltonian system, then the function $\hat{H}$ defined by

$$\hat{H} \equiv H + \{H, G\} + \frac{1}{2!} \{\{H, G\}, G\} + \frac{1}{3!} \{\{\{H, G\}, G\}, G\} + \dots, \quad (16)$$

is the result of applying a canonical change to H . This change is the time one flow corresponding to the Hamiltonian G . G is usually called the generating function of the transformation (16). See [20] and references therein for more details.

It is easy to check that, if P and Q are two homogeneous polynomials of degree r and s respectively, then $\{P, Q\}$ is a homogeneous polynomial of degree $r + s - 2$. This property is very useful to implement in a computer the transformation (16): for instance, let us assume that we want to eliminate the monomials of degree 3 of (15), as it is usually done in a normal form scheme. Let us select as a generating function a homogeneous polynomial of degree 3, G_3 . Then, it is immediate to check that the terms of $\hat{H}$ satisfy

degree 2: $\hat{H}_2 = H_2$,

degree 3: $\hat{H}_3 = H_3 + \{H_2, G_3\}$,

degree 4: $\hat{H}_4 = H_4 + \{H_3, G_3\} + \frac{1}{2!} \{\{H_2, G_3\}, G_3\}$,

$\vdots$

Hence, to kill the monomials of degree 3 one has to look for a G_3 such that $\{H_2, G_3\} = -H_3$. Let us denote

$$H_3(q, p) = \sum_{|k_q|+|k_p|=3} h_{k_q, k_p} q^{k_q} p^{k_p}, \quad G_3(q, p) = \sum_{|k_q|+|k_p|=3} g_{k_q, k_p} q^{k_q} p^{k_p},$$

and $H_2(q, p) = \sum_{j=1}^3 \eta_j q_j p_j$, where $\eta_1 = \lambda_1$, $\eta_2 = \sqrt{-1}\omega_1$ and $\eta_3 = \sqrt{-1}\omega_2$. As

$$\{H_2, G_3\} = \sum_{|k_q|+|k_p|=3} \langle k_p - k_q, \eta \rangle g_{k_q, k_p} q^{k_q} p^{k_p}, \quad \eta = (\eta_1, \eta_2, \eta_3),$$

it is immediate to obtain

$$G_3(q, p) = \sum_{|k_q|+|k_p|=3} \frac{-h_{k_q, k_p}}{\langle k_p - k_q, \eta \rangle} q^{k_q} p^{k_p},$$

that is well defined because the condition $|k_q| + |k_p| = 3$ implies that $\langle k_p - k_q, \eta \rangle$ is different from zero. Note that G_3 is so easily obtained because of the ‘diagonal’ form of H_2 given in (13).

In this paper we are not interested in a complete normal form, but only in uncoupling the central directions from the hyperbolic one. Hence, it is not necessary to cancel all the monomials in H_3 but only some of them. Moreover, as we want the radius of convergence of the transformed Hamiltonian to be as big as possible, we will try to choose the change of variables as close to the identity as possible. This means that we will kill the least possible number of monomials in the Hamiltonian. In order to produce an approximate first integral having the center manifold as a level surface (see below), it is enough to kill the monomials $q^{k_q} p^{k_p}$ such that the first component of k_q is different from the first component of k_p (other possibilities are discussed in [21]). This implies that the generating function G_3 is

$$G_3(q, p) = \sum_{(k_q, k_p) \in \mathcal{S}_3} \frac{-h_{k_q, k_p}}{\langle k_p - k_q, \eta \rangle} q^{k_q} p^{k_p}, \quad (17)$$

where $\mathcal{S}_n$, $n \geq 3$, is the set of indices (k_q, k_p) such that $|k_q| + |k_p| = n$ and the first component of k_q is different from the first component of k_p . Then, the transformed Hamiltonian $\hat{H}$ takes the form

$$\hat{H}(q, p) = H_2(q, p) + \hat{H}_3(q, p) + \hat{H}_4(q, p) + \cdots, \quad (18)$$

where $\hat{H}_3(q, p) \equiv \hat{H}_3(q_1 p_1, q_2, p_2, q_3, p_3)$ (note that $\hat{H}_3$ depends on the product $q_1 p_1$, not on each variable separately). This process can be carried out up to a finite order N , to obtain a Hamiltonian of the form

$$\bar{H}(q, p) = \bar{H}_N(q, p) + R_N(q, p),$$

where $H_N(q, p) \equiv H_N(q_1 p_1, q_2, p_2, q_3, p_3)$ is a polynomial of degree N and R_N is a remainder of order $N + 1$ (note that H_N depends on the product $q_1 p_1$ while the remainder depends on the two variables q_1 and p_1 separately). Now, neglecting the remainder and applying the canonical change given by $I_1 = q_1 p_1$, we obtain the Hamiltonian $\bar{H}_N(I_1, q_2, p_2, q_3, p_3)$ that has I_1 as a first integral. Then, setting $I_1 = 0$ (this is to skip the hyperbolic behavior) we obtain a two degrees of freedom Hamiltonian, $\bar{H}_N(0, \bar{q}, \bar{p})$, $\bar{q} = (q_2, q_3)$, $\bar{p} = (p_2, p_3)$, that represents (up to some finite order N) the dynamics inside the center manifold. Finally, the Hamiltonian is put into its real form by using the inverse of (12).

It is interesting to note the absence of small divisors during the entire process. The denominators that appear in the generating functions (like (17)), $\langle k_p - k_q, \eta \rangle$, can be bounded from below when $(k_q, k_p) \in \mathcal{S}_N$: using that η_1 is real and that $\eta_{2,3}$ are purely imaginary, we have

$$|\langle k_p - k_q, \eta \rangle| \geq |\lambda_1|, \quad \text{for all } (k_q, k_p) \in \mathcal{S}_N, \quad N \geq 3.$$

For this reason, the divergence of this process is very mild (for a discussion of this phenomenon in this context, see [21]; the divergence of normalizing transformations in the absence of small divisors has been considered in other contexts, see, for instance, [22] or [23]). This is clearly observed when this process is stopped at some degree N . Then, the remainder is very small in a quite big neighborhood of the equilibrium point. We will deal with these points in the next section.

An explicit expression for the change of variables that goes from the coordinates of the center manifold to the coordinates corresponding to Hamiltonian (15) can be obtained in the following way: once the generating function G_3 has been obtained, we can compute

$$\tilde{q}_j = q_j + \{q_j, G_3\} + \frac{1}{2!} \{\{q_j, G_3\}, G_3\} + \frac{1}{3!} \{\{\{q_j, G_3\}, G_3\}, G_3\} + \cdots, \quad (19)$$

$$\tilde{p}_j = p_j + \{p_j, G_3\} + \frac{1}{2!} \{\{p_j, G_3\}, G_3\} + \frac{1}{3!} \{\{\{p_j, G_3\}, G_3\}, G_3\} + \cdots, \quad (20)$$

that produces the transformation that sends the coordinates of (15), given by the variables $(\tilde{q}, \tilde{p})$ into the coordinates of (18), represented by the variables (q, p) . In the next step, the generating function G_4 is applied to the right-hand side of Eqs. (19) and (20), to obtain the change corresponding to fourth order, and so on. Then, substituting $q_1 = p_1 = 0$ one obtains six power expansions (corresponding to the six initial variables), each one depending on the four variables of the center manifold. Finally, these expansions are put into real form in the same way as the Hamiltonian.

From now on, and to simplify notation, the realified variables will still be called q and p .

Note that the normalizing method used here is slightly different than the one introduced in [24]. The main difference is that our process uses less computer memory, since it does not require to store the complete Lie triangle.

3.2. Numerical results

We have implemented the Lie series method to obtain the (approximate) center manifold for the equilibrium points $L_{1,2,3}$.

3.2.1. Software

As the commercial algebraic manipulators are not efficient enough to deal with big expansions, we have written our own software from scratch, using C language. Full details of the software (as well as the source code) can be found in [14].

The software has been run on a Linux PC (using the GNU compiler gcc/g++, version 2.7.2) with a 200 MHz. Pentium Pro CPU. Each center manifold computation, up to order 32, took about 15h 30m and 38MB of RAM memory. Note that a single expansion like (4) contains 1 388 577 monomials.

3.2.2. The L_1 point of the Earth–Sun system

We have applied the above-explained algorithm to the collinear points of the RTBP. As a first example we have focused on the L_1 point corresponding to the mass parameter $\mu = 3.0404233984441761 \times 10^{-6}$. This is an approximate value for the Earth–Sun case. All the expansions have been performed up to degree $N = 32$.

The first terms of the Hamiltonian restricted to the center manifold are in Table 1. The monomials that are not listed there have zero coefficient, due to the symmetries of this problem. In order to have an idea of the radius of convergence of this series, we have computed (numerically) the values

$$r_n^{(1)} = \frac{\|H_n\|_1}{\|H_{n-1}\|_1}, \quad r_n^{(2)} = \sqrt[n]{\|H_n\|_1}, \quad \text{where } \|H_n\|_1 = \sum_{|k|=n} |h_k|, \quad 3 \leq n \leq N, \quad (21)$$

being h_k the coefficient of the monomial of exponent k . These values have been plotted in Fig. 4. They seem to show a mild divergence (of logarithmic type, see [21]) of the series, although the size of the region where the (truncated) series looks convergent is quite big. Numerical (and very realistic) estimates of the radius of convergence are obtained as follows: take an initial condition inside the center manifold and, by means of a numerical integration of the reduced Hamiltonian, produce a sequence of points for the corresponding trajectory. Then, by means of the change of variables, send these points back to the initial RTBP coordinates. Finally, by means of a numerical integration of the RTBP, we can test if those points belong to the same orbit (note that we can not use a very long time span for those integrations, since the hyperbolic character of the center manifold in the RTBP amplifies the errors exponentially). This gives an idea of the error we have in the determination of the center manifold. In fact,

Table 1

Coefficients, up to degree 5, of the transformed Hamiltonian restricted to the center manifold corresponding to the L_1 point of the Earth–Sun system. The exponents (k_1, k_2, k_3, k_4) refer to the variables (q_2, p_2, q_3, p_3) , in this order

k_1	k_2	k_3	k_4	h_k	k_1	k_2	k_3	k_4	h_k
2	0	0	0	1.0432267821115535e+00	0	0	2	2	1.2424817827573600e-01
0	2	0	0	1.0432267821115544e+00	4	1	0	0	-2.0023568581469642e-01
0	0	2	0	1.0076053314983200e+00	2	3	0	0	3.4353440405951968e-01
0	0	0	2	1.0076053314983200e+00	0	5	0	0	-2.0187593581785741e-02
2	1	0	0	6.5165140304211688e-01	2	1	2	0	-1.9849089558605101e-01
0	3	0	0	-4.1659670417917148e-02	0	3	2	0	1.4712780865620459e-01
0	1	2	0	5.3911539423589860e-01	0	1	4	0	-2.7451664895216100e-02
4	0	0	0	-8.5787309100706366e-02	3	0	1	1	-1.1415906236784655e-01
2	2	0	0	4.1161447802927803e-01	1	2	1	1	2.1573064571205472e-01
0	4	0	0	-2.6563655599297287e-02	1	0	3	1	-9.4058985172178297e-02
2	0	2	0	-1.4043712336878425e-01	2	1	0	2	1.9372724033920288e-01
0	2	2	0	2.7927960671292551e-01	0	3	0	2	-4.4040459096995777e-02
0	0	4	0	-5.7468618566454702e-02	0	1	2	2	1.9106055501181426e-01
1	1	1	1	6.2490402334472867e-02	1	0	1	3	3.8405228183256930e-02
2	0	0	2	1.5018398762952467e-01	0	1	0	4	-2.2759839111536957e-02
0	2	0	2	-2.8803507814853090e-02					

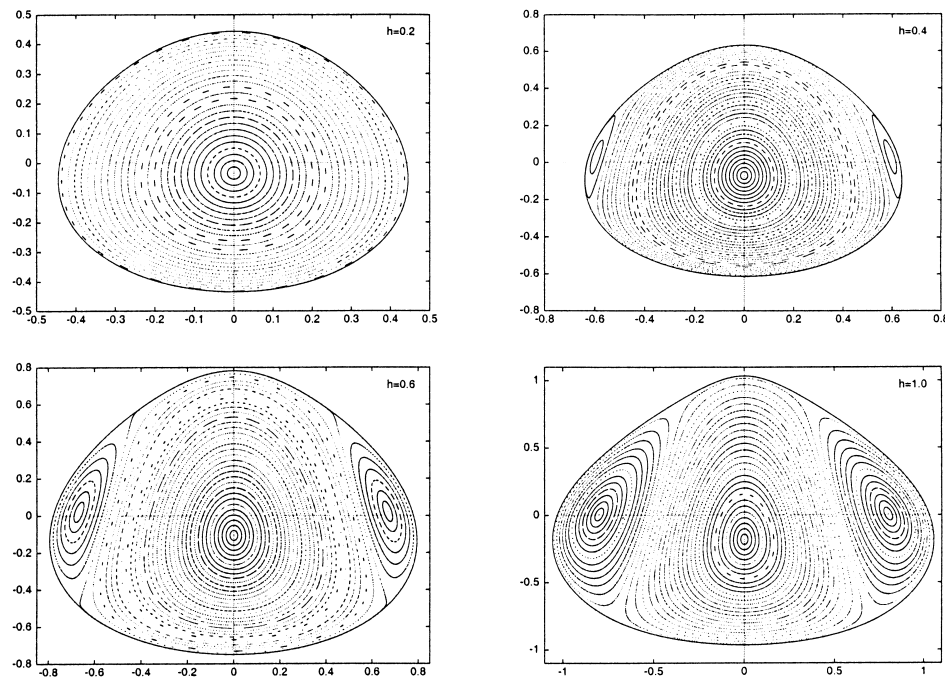


Fig. 3. Poincaré sections of the center manifold of L_1 , corresponding to $h = 0.2, 0.4, 0.6$ and 1.0 .

the accuracy of the plots in Figs. 3 and 6 has been checked in this way. For more details about this kind of error estimation, see [14].

To have a description of the dynamics inside the center manifold we use the following scheme: we take the 3D Poincaré section $q_3 = 0$ (this corresponds, at first order, to use $z = 0$ in the synodical coordinates) and we fix an energy level h_0 to obtain a 2D section. Hence, to obtain a picture of the phase space of this Poincaré section we select a value h_0 and an initial point (q_2, p_2) . Using that $q_3 = 0$ and that the value of the Hamiltonian must be

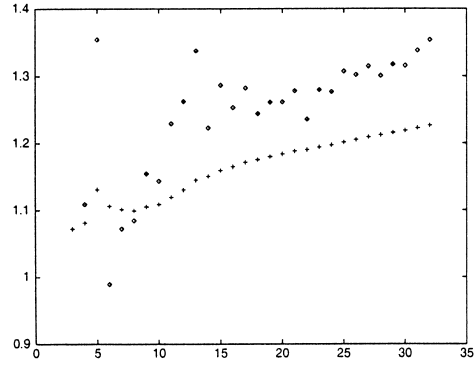


Fig. 4. Horizontal axis: values of n . Vertical axis: the corresponding values of $r_n^{(1)}$ ($\diamond$) and $r_n^{(2)}$ ($+$) according to (21), for the L_1 case.

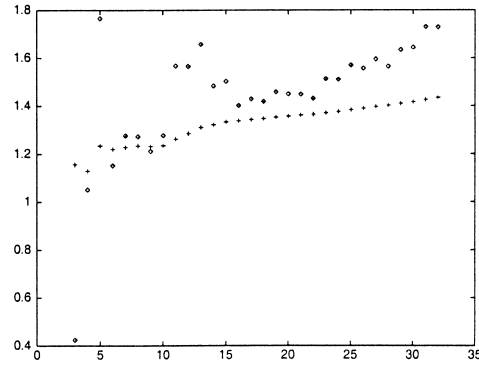


Fig. 5. Horizontal axis: values of n . Vertical axis: the corresponding values of $r_n^{(1)}$ ($\diamond$) and $r_n^{(2)}$ ($+$) according to (21), for the L_2 case.

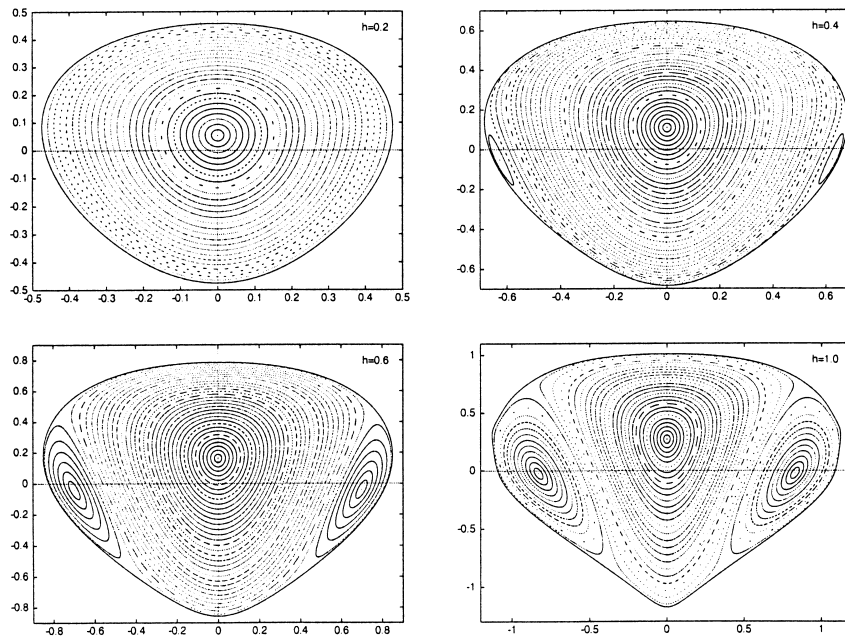


Fig. 6. Poincaré sections of the center manifold of L_2 , corresponding to $h = 0.2, 0.4, 0.6$ and 1.0 .

Table 2

Coefficients, up to degree 5, of the Hamiltonian restricted to the center manifold corresponding to the L_2 point of the Earth–Moon system. The exponents (k_1, k_2, k_3, k_4) refer to the variables (q_2, p_2, q_3, p_3) , in this order

k_1	k_2	k_3	k_4	h_k	k_1	k_2	k_3	k_4	h_k
2	0	0	0	9.3132294092164980e–01	0	0	2	2	1.6516656013507769e–01
0	2	0	0	9.3132294092164991e–01	4	1	0	0	3.0065634937222852e–01
0	0	2	0	8.9308808149867502e–01	2	3	0	0	–5.8388370855924443e–01
0	0	0	2	8.9308808149867525e–01	0	5	0	0	3.1707966658149511e–02
2	1	0	0	–8.3074621158508666e–01	2	1	2	0	2.4502956646982510e–01
0	3	0	0	6.5285116341699909e–02	0	3	2	0	–1.9424915041015245e–01
0	1	2	0	–6.4906335171207086e–01	0	1	4	0	8.2423593768551455e–03
4	0	0	0	–3.0986677967027330e–02	3	0	1	1	1.7854668840138077e–01
2	2	0	0	5.9388694902317307e–01	1	2	1	1	–4.2089157809945715e–01
0	4	0	0	–4.1582038336828324e–02	1	0	3	1	1.3900075305874218e–01
2	0	2	0	–4.7016550083469763e–02	2	1	0	2	–2.8565999503473249e–01
0	2	2	0	3.5694318621877408e–01	0	3	0	2	7.7193563104132376e–02
0	0	4	0	–1.7818840096908990e–02	0	1	2	2	–2.8562245235708378e–01
1	1	1	1	1.1056617867458479e–01	1	0	1	3	–7.5028466176853367e–02
2	0	0	2	2.1139923206390523e–01	0	1	0	4	4.3954249987303046e–02
0	2	0	2	–4.9839132339243322e–02					

h_0 , we compute (numerically) the corresponding value p_3 (in fact, there are two values that solve the equation, one positive and one negative; we use the positive one). Then, this point is used as initial condition for a numerical integration of the Hamiltonian restricted to the central part, plotting a point each time that the trajectory crosses the plane $q_3 = 0$ with $p_3 > 0$.

The results can be seen in Fig. 3. As the Hamiltonian is positive definite at the origin (this is clearly seen looking at the sign of the coefficients of the second degree terms in Table 1), each energy level defines a closed region in the Poincaré section. The boundary of this region coincides with a periodic orbit of the planar Lyapunov family of L_1 , that is fully contained in the plane $q_3 = p_3 = 0$, and in the figure it has been plotted using a continuous line. The motion inside this region is clearly quasi-periodic (except by some gaps that are too small to appear in these pictures), with a fixed point on the p_1 axis, that corresponds to a vertical Lyapunov orbit. If the value of the energy increases, one can see how the well known Halo orbits are born, as a bifurcation of the planar Lyapunov family. Note that the Halo orbits are surrounded by 2D invariant tori (see also [25] and [12]). The boundary between the tori around the Halo orbit and the tori around the vertical Lyapunov orbit is a homoclinic trajectory of the planar Lyapunov orbit. Of course, the homoclinic trajectory that goes out from the orbit and the one that goes in do not generally coincide: they should intersect one each other with a very small angle. This phenomenon is known as splitting of separatrices. Finally, by sending those orbits to the RTBP coordinates, it is possible to see that the description provided by those plots is valid (and very accurate) up to a distance of L_1 a little bit bigger than 60% of the L_1 -Earth distance.

3.2.3. The L_2 point of the Earth–Moon system

It is not difficult to repeat the computations of the last section for the L_2 point of the Earth–Moon system. The Hamiltonian restricted to the center manifold is displayed in Table 2, and the estimates of the region of convergence are shown in Fig. 5. Fig. 6 contains the plots of the Poincaré sections, where the bifurcation that gives rise to the Halo orbits is also shown. As before, it is possible to check that this description is very accurate up to a half distance from L_2 to the Moon. As the results are very similar to the last case, we do not add further remarks.

Table 3

Coefficients, up to degree 5, of the Hamiltonian restricted to the center manifold corresponding to the L_3 point of the Earth–Moon system. The exponents (k_1, k_2, k_3, k_4) refer to the variables (q_2, p_2, q_3, p_3) , in this order

k_1	k_2	k_3	k_4	h_k	k_1	k_2	k_3	k_4	h_k
2	0	0	0	5.0520994612145753e−01	0	0	2	2	7.1591406119636058e−01
0	2	0	0	5.0520994612145753e−01	4	1	0	0	2.1323011406157320e+03
0	0	2	0	5.0266571276387528e−01	2	3	0	0	−5.6249141438829110e+02
0	0	0	2	5.0266571276387528e−01	0	5	0	0	1.3204637130128873e+01
2	1	0	0	−5.6115912436382951e+00	2	1	2	0	5.2838311677563991e+02
0	3	0	0	9.3496383336128464e−01	0	3	2	0	−3.0623587048451135e+01
0	1	2	0	−1.4686056924068396e+00	0	1	4	0	1.5515423765627251e+01
4	0	0	0	7.6528931476095536e+00	3	0	1	1	1.8825389781246327e+00
2	2	0	0	1.3546730001510483e+01	1	2	1	1	−3.6872790343761423e+01
0	4	0	0	3.8069286794284551e−01	1	0	3	1	1.7599729630272192e+00
2	0	2	0	3.3499313360906959e+00	2	1	0	2	5.0848622203654816e+02
0	2	2	0	7.9659730977543119e−01	0	3	0	2	−2.9960960321274250e+01
0	0	4	0	3.5254807289843670e−01	0	1	2	2	2.7128291101879991e+01
1	1	1	1	4.0669264872473150e+00	1	0	1	3	−5.2867739300861096e+00
2	0	0	2	2.7355314621061146e+00	0	1	0	4	1.7913955620592070e+01
0	2	0	2	−1.3673249909233478e+00					

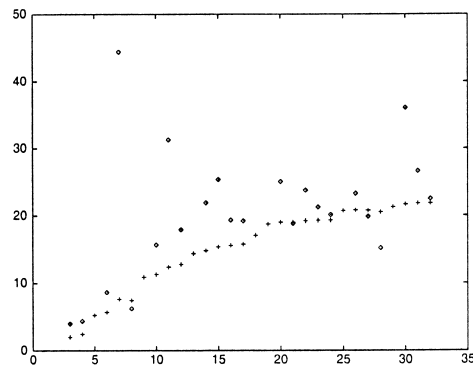


Fig. 7. Horizontal axis: values of n . Vertical axis: the corresponding values of $r_n^{(1)}$ ($\diamond$) and $r_n^{(2)}$ ($+$) according to ((21)), for the L_3 case.

3.2.4. The L_3 point of the Earth–Moon system

We have also performed these computations for the L_3 point. The main difference between this point and $L_{1,2}$ can be seen in Fig. 1: while $L_{1,2}$ are strongly influenced by the two bodies, L_3 is influenced by the biggest primary but the effect of the smallest primary is almost neglectable. This implies that the dynamics near L_3 is rather close to the dynamics of a two body problem (that is very degenerate). This is the reason for the big coefficients shown in Table 3, that are responsible for the poor convergence radius shown in Fig. 7. Hence, one must use very small values of the energy in order to be inside the region of convergence. This does not allow to go far enough to observe the bifurcation corresponding to the Halo orbits and, for this reason, we have not included the corresponding plots for this case.

So, the study of the behavior around L_3 (including the computation of Halo orbits) is a difficult problem. Moreover, as far as we know, there are no astronautical or astronomical applications that require an accurate knowledge of the phase space around this point. We believe that, in case that this study were necessary, it would be better to look at it as a slightly perturbed two body problem rather than using the techniques discussed here.

4. Semi-analytical computation of invariant tori – the Lindstedt–Poincaré method

Here we focus on the computation of periodic and quasi-periodic solutions contained in the center manifold. As we want to avoid using changes of variables, we will work directly on Eq. (14).

4.1. Lissajous Orbits

We follow the idea of the Lindstedt–Poincaré method to look for these 2-D invariant tori as a series expansion in two frequencies, which formally satisfies the equations of motion up to a selected order (to be defined later). The procedure, as we will see, is a recursion that determines the coefficients of each order from the ones corresponding to lower orders that have been computed in previous steps.

As a starting point we need the (non trivial) librating solutions of the linear part of (14),

$$\ddot{x} - 2\ddot{y} - (1 + 2c_2)x = 0, \quad \ddot{y} + 2\ddot{x} + (c_2 - 1)y = 0, \quad \ddot{z} + c_2z = 0, \quad (22)$$

that can be written as

$$x(t) = \alpha \cos(\omega_0 t + \phi_1), \quad y(t) = \kappa \alpha \cos(\omega_0 t + \phi_1), \quad z(t) = \beta \cos(\nu_0 t + \phi_2),$$

where

$$\omega_0 = \sqrt{\frac{2 - c_2 + \sqrt{9c_2^2 - 8c_2}}{2}}, \quad \nu_0 = \sqrt{c_2} \quad \text{and} \quad \kappa = \frac{-(\omega_0^2 + 1 + 2c_2)}{2\omega_0}$$

(see Section 2.1). The free parameters α and β , usually called in plane and out of plane amplitudes respectively, and ϕ_1, ϕ_2 , from now on called the phases, give all the librating solutions of the linear part of the equations.

We note that, due to the autonomous character of the equations, only one phase is needed. Hence, a solution is determined by three parameters, α, β and a phase. Nevertheless we will keep two phases because, as we will see, the computations in both cases are exactly the same.

When we consider the nonlinear terms of the equations, we look for formal expansions in powers of the amplitudes α and β , of the type

$$\begin{aligned} x(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i, |m| \leq j} x_{ijkm} \cos(k\theta_1 + m\theta_2) \right) \alpha^i \beta^j, & y(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i, |m| \leq j} y_{ijkm} \sin(k\theta_1 + m\theta_2) \right) \alpha^i \beta^j, \\ z(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i, |m| \leq j} z_{ijkm} \cos(k\theta_1 + m\theta_2) \right) \alpha^i \beta^j, \end{aligned}$$

where $\theta_1 = \omega t + \phi_1$ and $\theta_2 = \nu t + \phi_2$. Due to the presence of nonlinear terms the frequencies ω and ν can be kept no longer constant and they must be expanded as well in powers of the amplitudes: $\omega = \sum_{i,j=0}^{\infty} \omega_{ij} \alpha^i \beta^j$ and $\nu = \sum_{i,j=0}^{\infty} \nu_{ij} \alpha^i \beta^j$.

The goal is to compute the coefficients $x_{ijkm}, y_{ijkm}, z_{ijkm}, \omega_{ij}$ and ν_{ij} iteratively up to a finite order $i + j = n_f$ with the following meaning. Identifying the first coefficients of the general solution with the ones obtained from the solution of the linear part, we see that the non zero values are $x_{1010} = 1, y_{1010} = \kappa, z_{0101} = 1, \omega_{00} = \omega_0$ and $\nu_{00} = \nu_0$. The insertion of the expression of the solution of the linear part in (14) produces a remainder which are

series in α, β , beginning with terms of order $i + j = 2$. In what follows, it is said that the solution is computed up to order $i + j = n$ when, after inserting the obtained expression for the solution into the equations, the remainder consists of three series beginning with terms of order $i + j = n + 1$.

Before going into the details of the computations we note that $x(t)$ and $z(t)$ are both written as a cosinus series and $y(t)$ as a sinus one. This can be done due to the symmetries of the problem and to the selected expression for the solution of the linear part. It is not difficult to see that if j is odd the coefficients x_{ijkm} and y_{ijkm} are zero, and if j is even the coefficients z_{ijkm} are zero. As it is usual in the Lindstedt–Poincaré method, we only work with coefficients such that $|k| \leq i$ and $|m| \leq j$ and, moreover, k and m must have the same parity as i and j respectively. Due to the symmetries of sinus and cosinus, we can assume that $k \geq 0$ and, when $k = 0$, we can assume $m \geq 0$. Finally, in the expansions of the frequencies ω and ν only terms with both i and j even appear. All these properties have been taken into account to save computing time and storage.

At the first step of the procedure we start with the series $x(t)$, $y(t)$ and $z(t)$ determined up to order one and the series ω and ν up to order zero. At a certain step we will have the solution up to order $n - 1$, this is, $x(t)$, $y(t)$ and $z(t)$ determined up to order $n - 1$ and ω and ν up to order $n - 2$ if $n - 1$ is odd, or up to order $n - 3$ if $n - 1$ is even. In any case, we will say that ω and ν are determined up to order $n - 2$, remembering that for odd orders the corresponding coefficients are zero.

Assume that we have computed the solution up to order $n - 1$. When we insert this expression in the right-hand side of the equations (14), we obtain three series determined up to order n . Denote by p , q and r these series, that are respectively of the same type as x , y and z . We are interested in finding the terms of order n of the solution by equating the (known) order n terms of p , q and r with the corresponding (and unknown) terms coming from the left hand side of Eq. (14). From now on (unless otherwise stated), and to simplify the notation, v_{ijkm} will refer to the terms with $i + j = n$ of a series v . When k and m are omitted it will refer to a frequency type series.

Hence, the order n of the equations must be equated from both sides of (14). In the left-hand side there appear some derivatives which can be computed in the following way,

$$\dot{x} = \frac{\partial x}{\partial \theta_1} \frac{d\theta_1}{dt} + \frac{\partial x}{\partial \theta_2} \frac{d\theta_2}{dt} = \omega \frac{\partial x}{\partial \theta_1} + \nu \frac{\partial x}{\partial \theta_2}, \quad \ddot{x} = \omega^2 \frac{\partial^2 x}{\partial \theta_1^2} + 2\omega\nu \frac{\partial^2 x}{\partial \theta_1 \partial \theta_2} + \nu^2 \frac{\partial^2 x}{\partial \theta_2^2}.$$

Similar expressions can be derived for $\dot{y}$, $\ddot{y}$, and $\ddot{z}$.

Each summand of the former expressions contains products of series of different type, for instance $\omega(\partial x / \partial \theta_1)$ in the first order derivatives and $\omega^2(\partial^2 x / \partial \theta_1^2)$ in the second order ones. Since the series x , y and z are known up to order $n - 1$ and ω and ν up to order $n - 2$, the order n of these products of series contains a known part which can be added to p_{ijkm} , q_{ijkm} , r_{ijkm} and an unknown part that consists of products of terms which have to be determined. Let us see this in more detail.

Let fv be one of the terms that appear in the computations of the first derivative (f denotes the series of a frequency and v denotes the derivative of a coordinate series). The known part of order n of fv is obtained multiplying the terms of order i_f of f with the terms of order j_v of v such that $i_f + j_v = n$, $i_f = 1, \dots, n - 2$. The unknown part of order n appears when multiplying the respective parts of order 0 and n , and order $n - 1$ and 1 of f and v . The following table summarizes the unknown parts of order n in the computations of the first derivatives of x and y . The symbol δ stands for Kronecker's delta.

f	v	$\omega(\partial x / \partial \theta_1)$	$\nu(\partial x / \partial \theta_2)$	$\omega(\partial y / \partial \theta_1)$	$\nu(\partial y / \partial \theta_2)$
0	n	$-\omega_{00}kx_{ijkm}$	$-\nu_{00}mx_{ijkm}$	$\omega_{00}ky_{ijkm}$	$\nu_{00}my_{ijkm}$
$n - 1$	1	$-\omega_{i-1j}\delta_{1k}\delta_{0m}$	0	$\kappa\omega_{i-1j}\delta_{1k}\delta_{0m}$	0

If f is a frequency-type series and v a coordinate series (x , y or z), the following table summarizes the unknown parts of order n in the computations of the second derivatives.

f	$\partial^2 v / \partial \theta^2$	$\omega^2 (\partial^2 v / \partial \theta_1^2)$	$-2\omega v (\partial^2 v / \partial \theta_1 \partial \theta_2)$	$v^2 (\partial^2 v / \partial \theta_2^2)$
0	n	$-\omega_{00}^2 k^2 v_{ijkm}$	$-2\omega_{00} v_{00} k m v_{ijkm}$	$-v_{00} m^2 v_{ijkm}$
$n-1$	1	$-f_{i-1j} v_{1010} \delta_{1k} \delta_{0m}$	0	$-f_{ij-1} v_{0101} \delta_{0k} \delta_{1m}$

The known parts of order n coming from the first and second derivatives of the left-hand side of Eq. (14) are added to p_{ijkm} , q_{ijkm} and r_{ijkm} . Let us denote by $\bar{p}_{ijkm}$, $\bar{q}_{ijkm}$ and $\bar{r}_{ijkm}$ the resulting values.

Let us give some details about the computation of the terms f_{i-1j} and f_{ij-1} of the former table. First, we note that both are terms of order $n-1$. They consist of an unknown part coming from the product of order zero of one series against order $n-1$ of the another, plus a known part coming from the remaining products that produce order $n-1$. As in the previous table f_{i-1j} represents the terms of order $n-1$ of ω^2 and f_{ij-1} represents the terms of order $n-1$ of v^2 , we can write $f_{i-1j} = 2\omega_{00}\omega_{i-1j} + \Omega_{i-1j}$ and $f_{ij-1} = 2v_{00}v_{ij-1} + N_{ij-1}$, where Ω_{i-1j} and N_{ij-1} stand for the known parts of order $n-1$ we have just mentioned.

So, the linear system of equations for the unknown terms of order n can be written as,

$$\begin{aligned}
 & -\left(\varpi_{km}^2 + 1 + 2c_2\right) x_{ijkm} - 2\varpi_{km} y_{ijkm} - 2(\omega_{00} + \kappa) \omega_{i-1j} \delta_{1k} \delta_{0m} = \bar{p}_{ijkm}, \\
 & -2\varpi_{km} x_{ijkm} + \left(c_2 - 1 - \varpi_{km}^2\right) y_{ijkm} - 2(\kappa \omega_{00} + 1) \omega_{i-1j} \delta_{1k} \delta_{0m} = \bar{q}_{ijkm}, \\
 & (c_2 - \varpi_{km}^2) z_{ijkm} - 2v_{00} v_{ij-1} \delta_{0k} \delta_{1m} = \bar{r}_{ijkm},
 \end{aligned} \tag{23}$$

where $\varpi_{km} = k\omega_{00} + mv_{00}$, $\bar{p}_{ijkm} = \bar{p}_{ijkm} + \Omega_{i-1j} \delta_{1k} \delta_{0m}$, $\bar{q}_{ijkm} = \bar{q}_{ijkm} + \kappa \Omega_{i-1j} \delta_{1k} \delta_{0m}$ and $\bar{r}_{ijkm} = \bar{r}_{ijkm} + N_{ij-1} \delta_{0k} \delta_{1m}$.

When $(k, m) \neq (1, 0)$ and $(k, m) \neq (0, 1)$ we can solve (23) to find x_{ijkm} , y_{ijkm} and z_{ijkm} . Note that, if the frequencies of the linearized system are non-resonant, the determinant of the matrix of this system is always different from zero, although it can be very small due to the small divisors problem. As usual, this problem becomes more relevant for high orders.

When $(k, m) = (1, 0)$ (this can only happen when n is odd) the determinant of the x - y part in the first two equations of (23) is zero. So, we normalize taking $x_{ij10} = 0$ and we can solve for y_{ij10} , ω_{i-1j} . Finally z_{ij10} is determined solving $(c_2 - \omega_{00}^2) z_{ij10} = \bar{p}_{ij10}$.

When $(k, m) = (0, 1)$ we can determine x_{ij01} and y_{ij01} by solving the first two equations of (23). As in the third equation the coefficient of z_{ij01} is zero, we normalize the solution taking $z_{ij01} = 0$ and solving $-2v_{00} v_{ij-1} = \bar{p}_{ij01} + N_{ij-1}$ for v_{ij-1} .

4.2. Halo orbits

Halo orbits are periodic orbits which bifurcate from the planar Lyapunov periodic orbits when the in plane (or intrinsic) and out of plane (or normal) frequencies are equal. This is a 1:1 resonance that appear as a consequence of the nonlinear terms of the equations and, hence, we have to look for these 1-D invariant tori as series expansion with a single frequency.

As Halo orbits are due to the nonlinear terms of the equations, they do not appear in the linearized Eq. (22). In order to apply the Lindstedt–Poincaré procedure we modify the equations of motion (14) by adding the product of

the factors Δ and z to the third equation,

$$\begin{aligned} \ddot{x} - 2\dot{y} - (1 + 2c_2)x &= \sum_{n \geq 2} c_{n+1}(n+1)T_n, & \ddot{y} + 2\dot{x} + (c_2 - 1)y &= y \sum_{n \geq 2} c_{n+1}R_{n-1}, \\ \ddot{z} + c_2z &= z \sum_{n \geq 2} c_{n+1}R_{n-1} + \Delta z. \end{aligned} \quad (24)$$

We will look for 1-D invariant tori of these equations, with the condition $\Delta = 0$. In the procedure, the factor Δ is expanded as a frequency-type series, this is, $\Delta = \sum_{i,j=0}^{\infty} d_{ij} \alpha^i \beta^j$. The coefficients d_{ij} will be computed iteratively.

We start looking for the (non trivial) librating solutions with one frequency of the linear part of (24),

$$\ddot{x} - 2\dot{y} - (1 + 2c_2)x = 0, \quad \ddot{y} + 2\dot{x} + (c_2 - 1)y = 0, \quad \ddot{z} + c_2z = d_{00}z.$$

It is not difficult to see that they can be written as

$$x(t) = \alpha \cos(\omega_0 t + \phi), \quad y(t) = \kappa \alpha \cos(\omega_0 t + \phi), \quad z(t) = \beta \cos(\omega_0 t + \phi),$$

with $d_{00} = c_2 - \omega_0^2$ and where

$$\omega_0 = \sqrt{2 - c_2 + \sqrt{9c_2^2 - 8c_2}/2}, \quad \kappa = \frac{-(\omega_0^2 + 1 + 2c_2)}{2\omega_0}$$

and ϕ is an arbitrary phase.

As in the case of Lissajous orbits, α and β are called the in plane and out of plane amplitudes respectively. Of course, Halo orbits depend only on one frequency or one amplitude since they are 1-D invariant tori. The relationship between α and β is contained in the condition $\Delta = 0$ which defines implicitly $\alpha = \alpha(\beta)$. We note that, at the current step, Halo orbits are determined up to order 1, and $\Delta = 0$ is read as $d_{00} = 0$, showing again that there are not Halo orbits in the linear part of the equations.

When we consider the nonlinear terms of (24), we look for formal expansions in powers of the amplitudes α and β of the type

$$\begin{aligned} x(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i+j} x_{ijk} \cos(k\theta) \right) \alpha^i \beta^j, & y(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i+j} y_{ijk} \sin(k\theta) \right) \alpha^i \beta^j, \\ z(t) &= \sum_{i,j=1}^{\infty} \left(\sum_{|k| \leq i+j} z_{ijk} \cos(k\theta) \right) \alpha^i \beta^j, \end{aligned}$$

where $\theta = \omega t + \phi$ and, as in the case of 2-D invariant tori, the frequency ω must be expanded as $\omega = \sum_{i,j=0}^{\infty} \omega_{ij} \alpha^i \beta^j$. Moreover, now we have the expansion of the constraint $\Delta = \sum_{i,j=0}^{\infty} d_{ij} \alpha^i \beta^j = 0$.

As in the computation of Lissajous orbits, the symmetries of the problem allow to only consider terms with j even for the series $x(t)$, $y(t)$ and with j odd for the series $z(t)$. Moreover, we can restrict ourselves to the case in which $0 \leq k \leq i + j$, having k and $i + j$ the same parity. Finally, the frequency series ω and Δ only contain terms with both i and j even.

The final goal is to compute the coefficients x_{ijk} , y_{ijk} , z_{ijk} , ω_{ij} and d_{ij} iteratively up to a finite order $i + j = n_f$, starting with the (pseudo) solution of the linear part whose non zero values are $x_{101} = 1$, $y_{101} = \kappa$, $z_{011} = 1$, $\omega_{00} = \omega_0$ and $d_{00} = c_2 - \omega_0^2$.

As it is usual in this kind of Lindstedt–Poincaré procedure, at some step we will have the series $x(t)$, $y(t)$ and $z(t)$ determined up to order $n - 1$ and the series ω and Δ up to order $n - 2$ (we recall that when $n - 1$ is even, the

terms of order $n - 2$ of these series are zero). When we insert these expansions in the right hand side of the Eqs. (24) we obtain the series p_{ijk} , q_{ijk} , r_{ijk} and Δz (note that the term Δz is not included in r_{ijk}). Of course, p_{ijk} , q_{ijk} and r_{ijk} are of the same type as x , y and z respectively. Note that they are determined up to order $i + j = n$, and that the product Δz is determined up to order $n - 2$ if $n - 1$ is odd, or up to order $n - 3$ if $n - 1$ is even.

Next, we have to compute the part of order n of the derivatives with respect to time, $\dot{x}$, $\dot{y}$, $\ddot{x}$, $\ddot{y}$ and $\ddot{z}$. As in the previous section, these computations involve products of series of different type such as $\dot{x} = \omega(\partial x / \partial \theta)$ or $\ddot{x} = \omega^2(\partial^2 x / \partial \theta^2)$. The same statement holds for the computation of the part of order n of Δz . As in the previous section, the part of order n of these products consists of an already known part and of an unknown one. The known part is added to the corresponding series p_{ijk} , q_{ijk} or r_{ijk} to obtain the new series $\bar{p}_{ijk}$, $\bar{q}_{ijk}$ and $\bar{r}_{ijk}$. The unknown terms are summarized in the following tables.

f	v	$\omega(\partial x / \partial \theta)$	$\omega(\partial y / \partial \theta)$	Δz
0	n	$-\omega_{00}kx_{ijk}$	$\omega_{00}ky_{ijk}$	$d_{00}z_{ijk}$
$n - 1$	1	$-\omega_{i-1j}\delta_{1k}$	$\omega_{i-1j}\kappa\delta_{1k}$	$d_{ij-1}\delta_{1k}$

f	$\partial^2 v / \partial \theta^2$	$\omega^2(\partial^2 x / \partial \theta^2)$	$\omega^2(\partial^2 y / \partial \theta^2)$	$\omega^2(\partial^2 z / \partial \theta^2)$
0	n	$-\omega_{00}^2k^2x_{ijk}$	$-\omega_{00}^2k^2y_{ijk}$	$-\omega_{00}^2k^2z_{ijk}$
$n - 1$	1	$-f_{i-1j}\delta_{1k}$	$-f_{i-1j}\kappa\delta_{1k}$	$-f_{ij-1}\delta_{1k}$

In the second table, f denotes the series corresponding to ω^2 . Note that its $n - 1$ order terms, f_{i-1j} and f_{ij-1} , are made of known and unknown parts, $f_{i-1j} = 2\omega_{00}\omega_{i-1j} + \Omega_{i-1j}$ and $f_{ij-1} = 2\omega_{00}\omega_{ij-1} + \Omega_{ij-1}$, where Ω_{i-1j} and Ω_{ij-1} denotes the known ones.

Finally, we can write the linear system of equations for the unknown terms of order n ,

$$\begin{aligned}
 & -\left(k^2\omega_{00}^2 + 1 + 2c_2\right)x_{ijk} - 2k\omega_{00}y_{ijk} - 2(\omega_{00} + \kappa)\omega_{i-1j}\delta_{1k} = \bar{p}_{ijk} + \Omega_{i-1j}\delta_{1k}, \\
 & -2k\omega_{00}x_{ijk} + \left(c_2 - 1 - k^2\omega_{00}^2\right)y_{ijk} - 2(\kappa\omega_{00} + 1)\omega_{i-1j}\delta_{1k} = \bar{q}_{ijk} + \kappa\Omega_{i-1j}\delta_{1k}, \\
 & \left(c_2 - k^2\omega_{00}^2 - d_{00}\right)z_{ijk} - d_{ij-1}\delta_{1k} = \bar{r}_{ijk} + \Omega_{ij-1}\delta_{1k} + 2\omega_{00}\omega_{ij-1}\delta_{1k}.
 \end{aligned} \tag{25}$$

When $k \neq 1$ we can solve (25) for x_{ijk} , y_{ijk} and z_{ijk} . When $k = 1$ the determinant of the xy part and the coefficient of z_{ijk} are zero. Then we can normalize by taking $x_{ij1} = 0$, $z_{ij1} = 0$, and solving the first two equations of (25) to find y_{ij1} and ω_{i-1j} . Finally the third equation is solved to obtain d_{ij-1} .

4.3. Numerical results

The algorithms presented for the semi-analytical computation of the Lissajous and Halo orbits have been implemented using Fortran 77 language. As in the computation of the center manifold, commercial algebraic manipulators are much less efficient dealing with expansions, specially in the case where symmetries can be efficiently implemented in the computations. The use of an ad hoc code allows us to reach very high orders (obtaining then very accurate solutions) in a short time. Table 4 summarizes the amount of RAM memory and CPU time used in some of these computations. Tables 5 and 6 show the coefficients of the Halo and Lissajous expansions around L_1 in the Earth–Sun system, up to order 3. Typical plots of Halo orbits, vertical periodic (Lyapunov) orbits and Lissajous trajectories are presented in Figs. 8–10, respectively. The accuracy of the expansions has been tested against numerical integration. For this purpose, initial conditions have been computed by tabulating the series expansions up

Table 4

Computer time (in seconds, for a Linux PC with a 200 MHz Pentium Pro CPU) and memory (in Kilobytes) for several degrees (N) of the Lindstedt–Poincaré computations

N	Halo		Lissajous	
15	113.2 Kb	<1 s	350.8 Kb	<1 s
17	171.7 Kb	<1 s	577.9 Kb	1 s
19	250.3 Kb	<1 s	909.3 Kb	2 s
21	353.2 Kb	<1 s	1377.2 Kb	5 s
23	484.8 Kb	1 s	2019.9 Kb	9 s
25	650.1 Kb	2 s	2882.1 Kb	16 s
27	854.2 Kb	3 s	4015.8 Kb	28 s
29	1103.0 Kb	6 s	5480.2 Kb	49 s
31	1402.4 Kb	9 s	7342.9 Kb	80 s
33	1758.9 Kb	13 s	9679.8 Kb	128 s
35	2179.2 Kb	19 s	12576.0 Kb	200 s
37	2670.6 Kb	27 s	16126.3 Kb	310 s
39	3240.7 Kb	37 s	20435.4 Kb	464 s
41	3897.4 Kb	51 s	25618.8 Kb	688 s
43	4649.1 Kb	70 s	31803.1 Kb	1003 s
45	5504.5 Kb	93 s	39126.5 Kb	1441 s

Table 5

Coefficients, up to order 3, of the Lindstedt–Poincaré expansion of the Halo orbits about L_1 in the Earth–Sun system (in this case, $\mu = 3.040423398444176E - 06$ and $\gamma = 0.1001097722778141E - 01$)

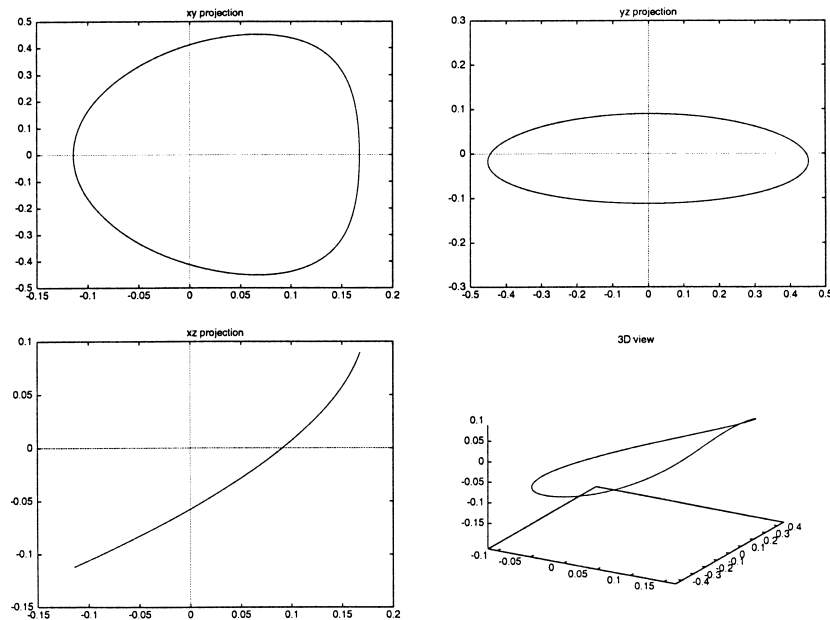
i	j		ω_{ij}	d_{ij}
0	0		.2086453564223108E + 01	-.2922144594039562E + 00
2	0		-.1720616528118310E + 01	.1596560311526045E + 02
0	2		.2526665927441598E + 00	-.1740900798763014E + 01
i	j	k	x_{ijk} or z_{ijk}	y_{ijk}
1	0	1	.1000000000000000E + 01	-.3229268251936296E + 01
0	1	1	.1000000000000000E + 01	
2	0	0	.2092695724506777E + 01	-.4778922922033039E + 01
2	0	2	-.9059648301914131E + 00	-.4924458783826867E + 00
0	2	0	.2482976576916406E + 00	.0000000000000000E + 00
0	2	2	.1044641085314726E + 00	-.6074645997077919E - 01
1	1	0	-.1040596322643552E + 01	
1	1	2	.3468654408811841E + 00	
3	0	1	.0000000000000000E + 00	.2845081624743493E + 01
3	0	3	-.7938202440824402E + 00	-.8857008912091187E + 00
1	2	1	.0000000000000000E + 00	.4316928130485082E + 00
1	2	3	.8268538161313826E - 01	.2301983693229456E - 01
2	1	3	.3980954407770784E + 00	
0	3	3	-.1904387085744166E - 01	

to different orders. Then, these initial conditions have been integrated (numerically) during π units of adimensional time. The final coordinates obtained from this numerical integration have been compared with the coordinates given by the series expansion at time π . If the difference in the euclidean norm for the positions is lower than 10^{-6} adimensional RTBP units we say that the initial condition is accurate (in the sense that it is very close to one of the periodic or quasi-periodic trajectories computed by the Lindstedt–Poincaré method). Similar results are obtained using euclidean norm with positions and velocities. The threshold has been selected to be 10^{-6} because, due to the instability of the orbits around the collinear points, an error of this amount in position coordinates after π units of

Table 6

Coefficients, up to order 3, of the Lindstedt–Poincaré expansion of the Lissajous orbits about L_1 in the Earth–Sun system

i	j			ω_{ij}	v_{ij}
0	0			.2086453564223108E+01	.2015210662996640E+01
2	0			−.1720616528118309E+01	.2227430750989766E+00
0	2			.2581841437578153E−01	−.1631915758176957E+00
i	j	k	m	x_{ijk} or z_{ijk}	y_{ijk}
1	0	1	0	.1000000000000000E+01	−.3229268251936296E+01
0	1	0	1	.1000000000000000E+01	
2	0	0	0	.2092695724506778E+01	−.4778922922033039E+01
2	0	2	0	−.9059648301914133E+00	−.4924458783826869E+00
0	2	0	0	.2482976576916407E+00	.0000000000000000E+00
0	2	0	2	.1108251822042930E+00	−.6776373426177420E−01
1	1	1	−1	−.1116868267568415E+01	
1	1	1	1	.3549452858304732E+00	
3	0	1	0	.0000000000000000E+00	.2845081624743493E+01
3	0	3	0	−.7938202440824405E+00	−.8857008912091185E+00
1	2	1	−2	−.1499994891576764E+01	−.4841968041750657E+01
1	2	1	0	.0000000000000000E+00	.2875532315811784E+00
1	2	1	2	.8387777659811270E−01	.2082881844639578E−01
2	1	2	−1	.1216565813734685E+02	
2	1	2	1	.4060793036977860E+00	
0	3	0	3	−.1952722175104363E−01	

Fig. 8. Projections on the coordinate planes and a 3D representation of a Halo orbit about L_1 in the Earth–Sun system with $\beta = 0.1$.

time, implies an error of about 100 meters in the initial conditions for the Earth–Sun system. This is because errors increase by a factor close to 1500 after π units of time due to the hyperbolic character of the orbits. Moreover, the use of different thresholds produces qualitatively similar results.

In this way, given a couple of amplitudes (α, β) , we can check if a mesh of points (initial conditions) on the corresponding Lindstedt–Poincaré series satisfy the accuracy criterion given above. In that case, we say that (α, β)

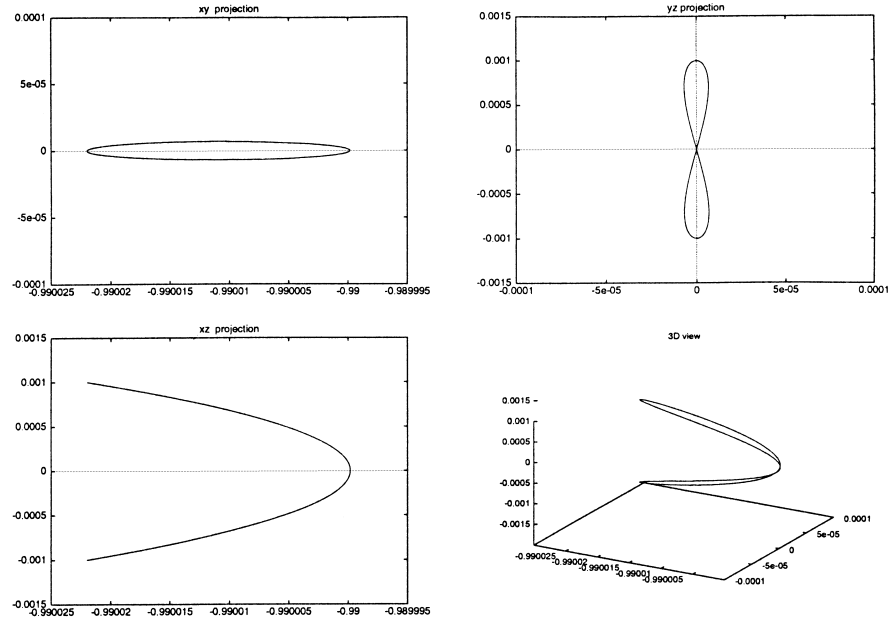


Fig. 9. Projections on the coordinate planes and a 3D representation of a vertical periodic orbit about L_1 in the Earth-Sun System and with $\alpha = 0.0$ and $\beta = 0.1$.

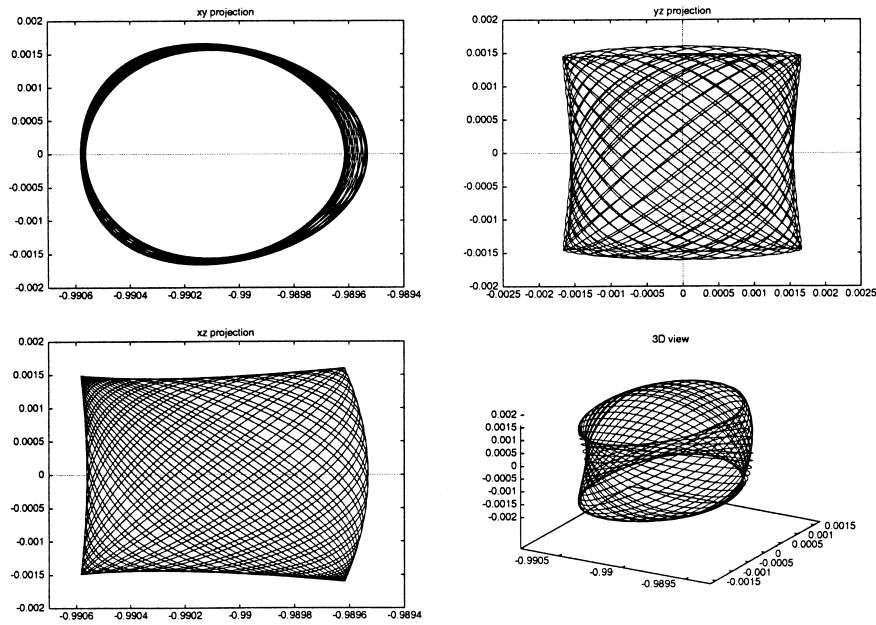


Fig. 10. Projections on the coordinate planes and a 3D representation of a Lissajous orbit about L_1 in the Earth-Sun System and with $\alpha = 0.05$ and $\beta = 0.15$.

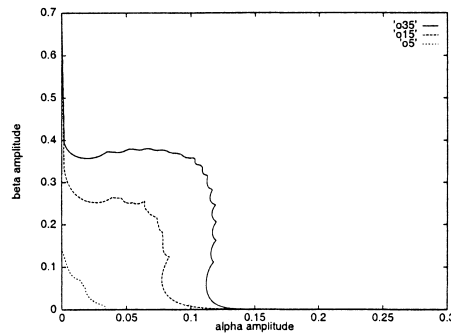


Fig. 11. Domain of practical convergence of the Lindstedt–Poincaré expansion of the Lissajous orbits around L_1 in the Earth–Sun system according to the criterion explained in the text. The different lines correspond to orders 5, 15 and 35.

Table 7

Maximum β amplitude which gives the region of convergence using the criterion explained in the text, for the case of Halo orbits around L_1 in the Earth–Sun system

Order	β amplitude
11	0.196
15	0.336
21	0.461
25	0.502
31	0.557
35	0.604

belongs to the domain of practical convergence. Here, ‘practical’ means that the series looks convergent for numerical purposes (in fact, it is well known that those series are divergent due to the effect of the small divisors).

The regions of practical convergence for the case of Lissajous orbits are represented in Fig. 11, using as example the expansion around L_1 for the Earth–Sun system. Each continuous line represents the boundary of the domain of practical convergence for a selected order of truncation of the expansion. Given any couple (α, β) inside the region of practical convergence, the error after the numerical integration is less than the selected threshold for all the phases corresponding to the different initial conditions. A similar plot is used and discussed in [11].

The same test has been done for the case of Halo orbits. In this case the expansion only depends on one amplitude. The maximum β amplitude reached for some orders is given in Table 7.

5. Conclusions

In this work, two different procedures have been implemented for describing and computing orbits inside the center manifold of the collinear points of the RTBP. The first method, named the reduction to the center manifold, gives a very good description of the dynamics inside the center manifold, but it does not produce closed formulas for the solutions, and it is quite expensive from the computational point of view. The second one, the Lindstedt–Poincaré method, gives ad-hoc formulae for the orbits and it runs quite fast in the computer, but the detailed global description is poor. The combination of both procedures gives a powerful tool to describe and compute solutions in an extended neighborhood of an equilibrium point. This methodology has been applied to study the dynamics around the collinear points of the RTBP, with special emphasis to the design of spacecraft trajectories.

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References

- [1] V. Szebehely, *Theory of Orbits*, Academic Press, New York, 1967.
- [2] L.H. Eliasson, Perturbations of stable invariant tori for Hamiltonian systems, *Ann. Sc. Norm. Super. Pisa Cl. Sci. Ser. IV* 15(1) (1988) 115–147.
- [3] À. Jorba, J. Villanueva, On the normal behaviour of partially elliptic lower dimensional tori of Hamiltonian systems, *Nonlinearity* 10 (1997) 783–822.
- [4] H. Poincaré, *Les Méthodes Nouvelles de la Mécanique Céleste*, Gauthier-Villars (1892, 1893, 1899).
- [5] R.W. Farquhar, Halo-orbit and Lunar-Swingby missions of the 1990's, *Acta Astronautica* 24 (1991) 227–234.
- [6] R.W. Farquhar, D.P. Muhonen, C. Newman, H. Heuberger, The first libration point satellite. Mission overview and flight history, *AAS/AIAA Astrodynamics Spec. Conf.*, 1979.
- [7] K.C. Howell, B.T. Barden, R.S. Wilson, M.W. Lo, Trajectory Design Using a Dynamical Systems Approach with Application to Genesis, *AAS Paper 97-709*. AAS/AIAA Astrodynamics Specialist Conference, Sun Valley, Idaho, 1997.
- [8] K.C. Howell, H.J. Pernicka, Numerical determination of Lissajous trajectories in the restricted three-body problem, *Celestial Mechanics* 41 (1988) 107–124.
- [9] M.C.E. Huber, The history of the SOHO mission, *ESA Bulletin* 86 (1996) 25–35.
- [10] D.L. Richardson, Halo orbit formulation for the ISEE-3 mission, *J. Guidance and Control* 3 (1980) 543–548.
- [11] G. Gómez, À. Jorba, J. Masdemont, C. Simó, Study Refinement of Semi-Analytical Halo Orbit Theory, *ESOC Contract 8625/89/D/MD(SC)*, Final Report, 1991.
- [12] G. Gómez, J. Masdemont, C. Simó, Quasihalo orbits associated with libration points, to appear in *Journal of the Astronautical Sciences*.
- [13] À. Jorba, J. Villanueva, On the persistence of lower dimensional invariant tori under quasi-periodic perturbations, *J. Nonlinear Science* 7 (1997) 427–473.
- [14] À. Jorba, A methodology for the numerical computation of normal forms, centre manifolds and first integrals of hamiltonian systems, to appear in *Exp. Math.*, 1999.
- [15] À. Jorba, J. Masdemont, Nonlinear dynamics in an extended neighbourhood of the translunar libration point, in: *Hamiltonian Systems with Three or More Degrees of Freedom*, C. Simó (Ed.), NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci. Kluwer, Dordrecht, Holland (to appear).
- [16] C. Simó, Effective computations in celestial mechanics and astrodynamics, lectures given at CISM Course on Modern Methods of Analytical Mechanics and their Applications, Udine, Italy, 1997.
- [17] J. Cobos, C. Simó, Reduction to the Central Manifold around a Collinear Libration Point of the Restricted Three Body Problem, *Proceedings of the XV CEDYA / V CMA*, 1998, pp. 673–676.
- [18] G. Gómez, J. Llibre, R. Martínez, C. Simó, Station Keeping of Libration Point Orbits, *ESOC Contract 5684/83/D/JS(SC)*, Final Report, 1985.
- [19] D.L. Richardson, A Note on a Lagrangian Formulation for Motion about the Collinear Points, *Celestial Mechanics* 22 (1980) 231–236.
- [20] A. Giorgilli, A. Delshams, E. Fontich, L. Galgani, C. Simó, Effective stability for a Hamiltonian system near an elliptic equilibrium point, with an application to the restricted three body problem, *J. Diff. Eq.* 77 (1989) 167–198.
- [21] À. Jorba, R. Llave, Regularity properties of center manifolds and applications, in preparation.
- [22] A. Bazzani, G. Giovannozzi, G. Servizi, E. Todesco, G. Turchetti, Resonant normal forms, interpolating Hamiltonians and stability analysis of area preserving maps, *Physica D* 64 (1993) 66–97.
- [23] G. Benettin, A. Giorgilli, On the Hamiltonian interpolation of near-to-the-identity symplectic mappings with application to symplectic integration algorithms, *J. Stat. Phys.* 74 (1994) 1117–1143.
- [24] A. Deprit, Canonical transformations depending on a small parameter, *Cel. Mech.* 1 (1969) 12–30.
- [25] G. Gómez, J. Masdemont, C. Simó, Lissajous orbits around Halo orbits, *Advances in the Astronautical Sciences* 95 (1997) 117–134.
- [26] À. Jorba, J. Villanueva, Numerical computation of normal forms around some periodic orbits of the restricted three body problem, *Physica D* 114 (1998) 197–229.